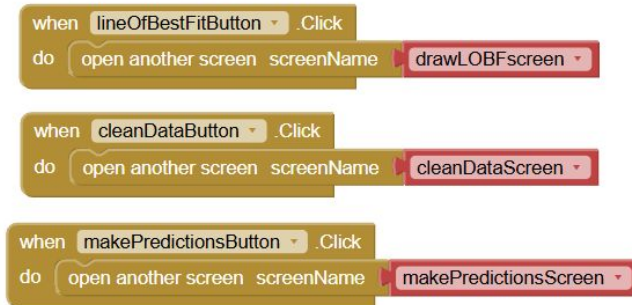
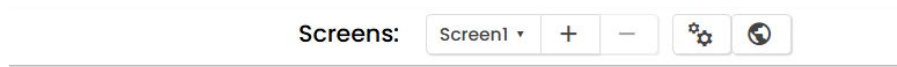


Camila es un joven que vive en Perú y planea viajar de vacaciones durante el verano, entre enero y marzo. Uno de los destinos que le interesa es Japón; sin embargo, sabe que es una zona altamente sísmica y le preocupa la posibilidad de que ocurra un terremoto durante su estadía. Camila quiere tomar una decisión informada y viajar con mayor tranquilidad, por lo que necesita conocer qué tan probable es que ocurran terremotos en Japón durante esos meses y si existe algún mes con menor frecuencia de eventos sísmicos. Para responder a esta inquietud, se analizarán datos históricos de terremotos registrados entre 2023 y 2025, con el fin de identificar patrones temporales que ayuden a comprender el riesgo sísmico durante el verano.

country	affected_population	response_time_hours	severity_level	estimated_economic_loss_usd
Japan	39988	63.28	7	7769396.31
Japan	51123	12.21	9	9368658.7
Japan	64557	54.37	10	15848097.56
Japan	0	44.8	1	0.0
Japan	40329	28.26	8	2646028.79
Japan	55402	4.39	10	2888787.85
Japan	44405	30.22	10	6542930.82
Japan	16762	13.31	4	2388828.2
Japan	49644	42.62	10	7398469.71
Japan	37214	35.16	7	9047672.04
Japan	9750	29.51	5	2254357.5
Japan	38050	69.04	7	5353005.45
Japan	14309	14.28	2	2452440.99
Japan	46588	23.22	9	3108795.61
Japan	15516	19.93	1	1618279.51
Japan	18772	61.84	2	2993149.04
Japan	23073	58.25	4	2172201.66
Japan	15239	66.99	2	2837501.65
Japan	32090	1.25	4	6044849.07
Japan	34404	65.25	5	6427751.59
Japan	29231	5.36	8	5394863.58
Japan	40411	10.09	10	4933510.83

Kaggle. (s.f.). *A Data-Driven Study of Natural Disaster Events* [Dataset]. Kaggle. Recuperado de <https://www.kaggle.com/datasets/minahilfatima12328/a-data-driven-study-of-natural-disaster-events/data>

Project Earthquake Risk: Analyzing Seismic Activity in Summer Months



Screens:

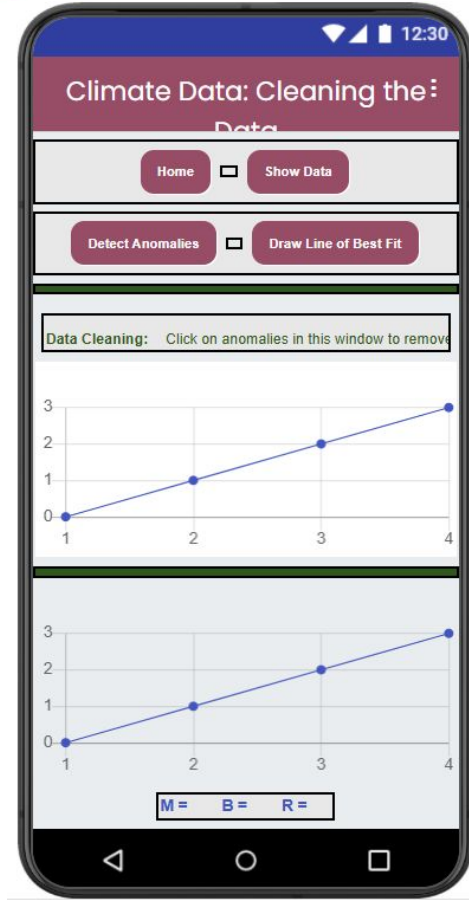
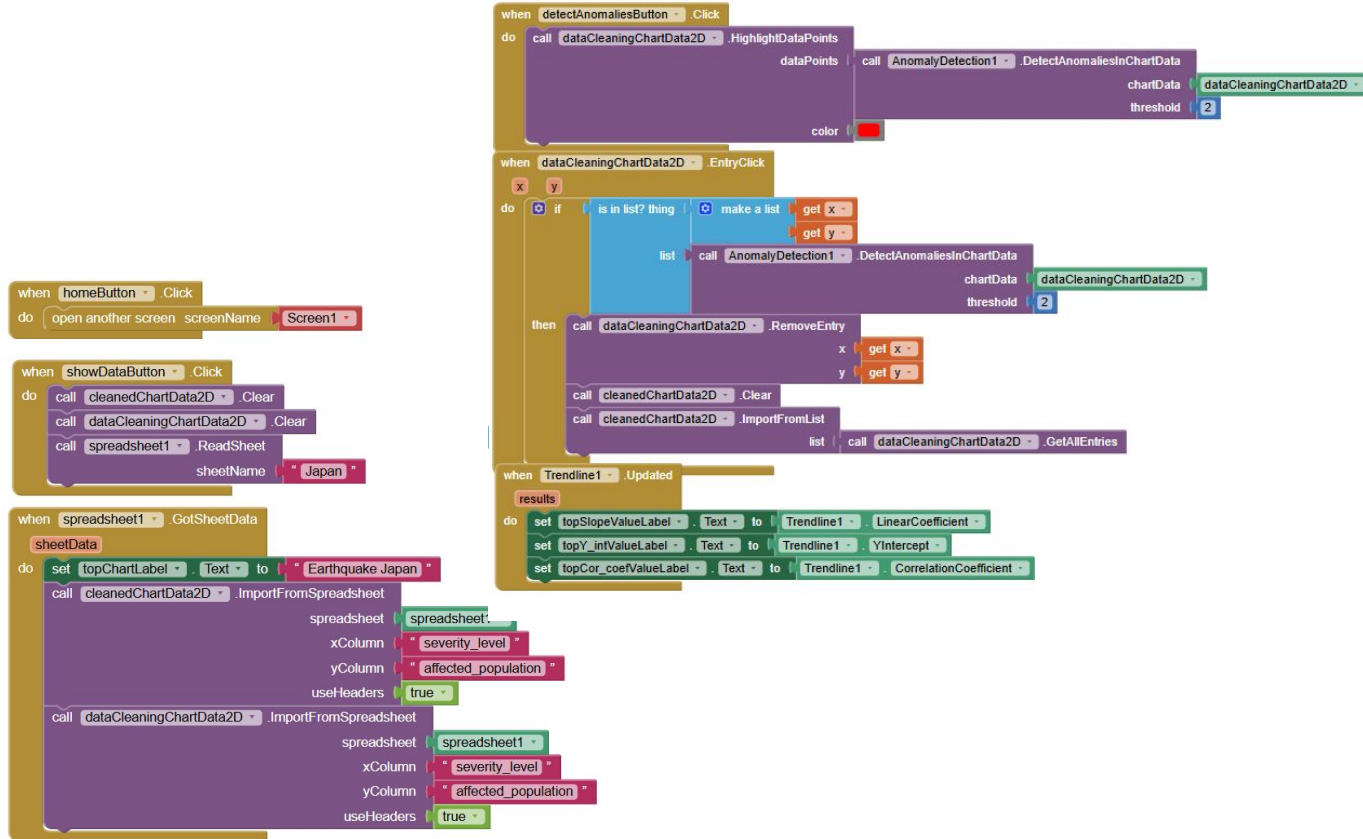
cleanDataScreen ▾

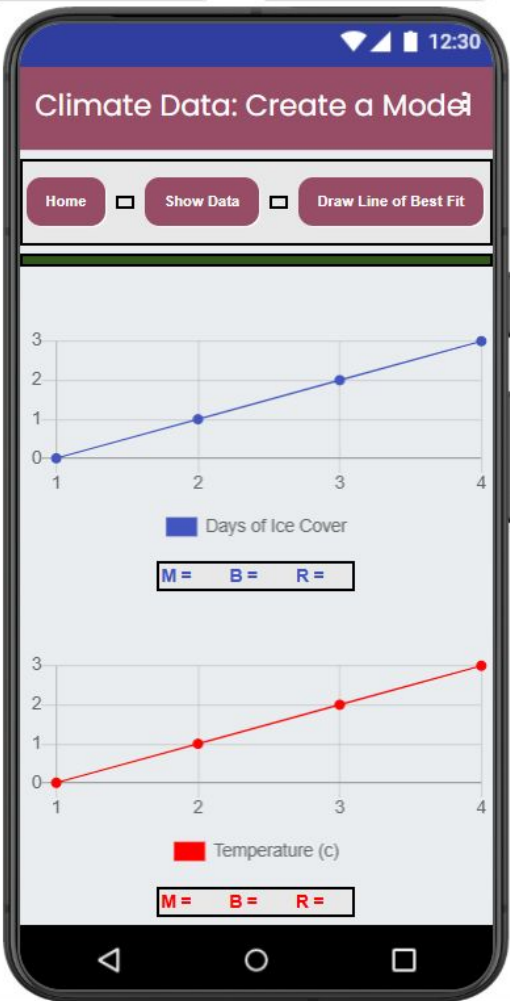
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```
when homeButton.Click
do open another screen screenName Screen1
```

```
when showDataButton.Click
do
  call topChartData2D.Clear
  call bottomChartData2D.Clear
  call spreadsheet1.ReadSheet
  sheetName Japan
```

```
when spreadsheet1.GotSheetData
sheetData
do
  set topChartLabel.Text to Earthquake Japan
  call topChartData2D.ImportFromSpreadsheet
  spreadsheet spreadsheet1
  xColumn severity_level
  yColumn affected_population
  useHeaders true
  call bottomChartData2D.ImportFromSpreadsheet
  spreadsheet spreadsheet1
  xColumn severity_level
  yColumn response_time_h
  useHeaders true
```

```
when drawLineOfBestFitButton.Click
do
  set topTrendline.ChartData to topChartData2D
  set bottomTrendline.ChartData to bottomChartData2D
  set topSlopeValueLabel.Text to topTrendline.LinearCoefficient
  set topY_intValueLabel.Text to topTrendline.YIntercept
  set topCor_coefValueLabel.Text to topTrendline.CorrelationCoefficient
  set bottomSlopeValueLabel.Text to bottomTrendline.LinearCoefficient
  set bottomY_intValueLabel.Text to bottomTrendline.YIntercept
  set bottomCor_coefValueLabel.Text to bottomTrendline.CorrelationCoefficient
```

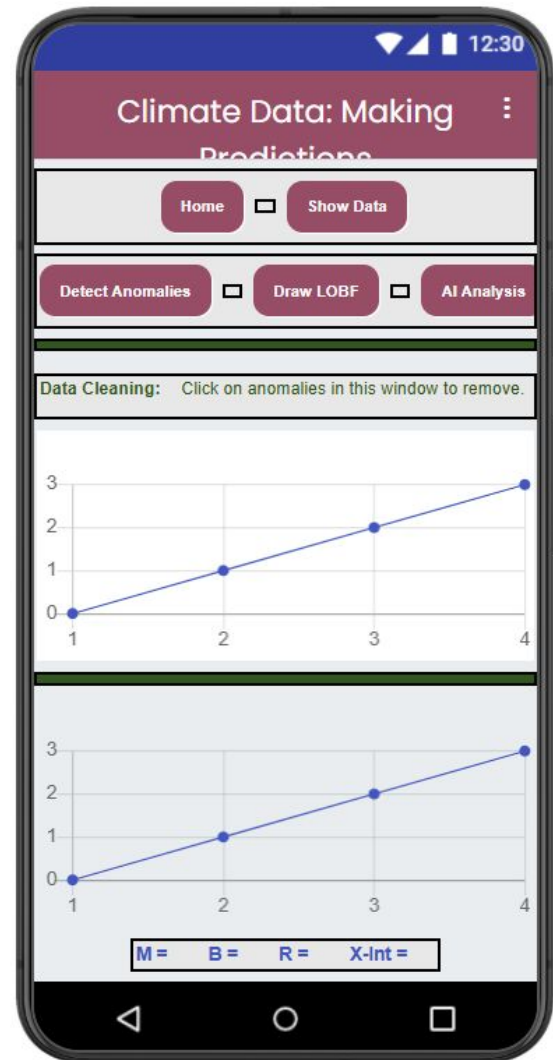
```

when homeButton .Click
do open another screen screenName Screen1

when showDataButton .Click
do call cleanedChartData2D .Clear
   call dataCleaningChartData2D .Clear
   call spreadsheet1 .ReadSheet
      sheetName "Japan"

when spreadsheet1 .GotSheetData
sheetData
do set topChartLabel .Text to "Earthquake Japan"
   call cleanedChartData2D .ImportFromSpreadsheet
      spreadsheet spreadsheet1
      xColumn "severity_level"
      yColumn "affected_population"
      useHeaders true
   call dataCleaningChartData2D .ImportFromSpreadsheet
      spreadsheet spreadsheet1
      xColumn "severity_level"
      yColumn "affected_population"
      useHeaders true

```




```

when drawLineOfBestFitButton . Click
do set Trendline1 . ChartData to cleanedChartData2D

```

```

when Trendline1 . Updated
results
do
  set SlopeValueLabel . Text to Trendline1 . LinearCoefficient
  set Y_intValueLabel . Text to round Trendline1 . YIntercept
  set Cor_coefValueLabel . Text to Trendline1 . CorrelationCoefficient
  set X_IntValueLabel . Text to round Trendline1 . XIntercepts
  call cleanedDataChart . ExtendDomainToInclude x Trendline1 . XIntercepts

```

```

when detectAnomaliesButton . Click
do
  call dataCleaningChartData2D . HighlightDataPoints
  dataPoints call AnomalyDetection1 . DetectAnomaliesInChartData
  chartData dataCleaningChartData2D
  threshold 2
  color

```

```

when AiAnalysisButton . Click
do
  set dataCleaningChart . Visible to false
  set dataCleaningHorizontalArrangement . Visible to false
  set AiResponseHorizontalArrangement . Visible to true
  call ChatBot1 . Converse
  question join
    " Datos de eventos de terremotos ocurridos entre e..."
    " Número de personas afectadas por terremotos. "
    call dataCleaningChartData2D . GetAllEntries
    " El coeficiente de correlación para la línea de m..."
    Trendline1 . CorrelationCoefficient
    " La pendiente de la línea de mejor ajuste es ..."
    Trendline1 . LinearCoefficient
    " La intersección en "Y" para la línea de mayor aj..."
    Trendline1 . YIntercept
    " Estimación de población afectada para un nivel e..."
    " ¿Cómo se relaciona esta tendencia con la intensi..."
    " ¿Qué pasa con las comunidades en zonas en sísmic..."
    " Limita tus respuestas a 120 palabras "

```

```

when dataCleaningChartData2D . EntryClick
x y
do
  call dataCleaningChartData2D . RemoveEntry
  x get x
  y get y
  call cleanedChartData2D . Clear
  call cleanedChartData2D . ImportFromList
  list call dataCleaningChartData2D . GetAllEntries

```

```

when ChatBot1 . GotResponse
responseText
do
  set AiResponseTextBox . Text to get responseText

```

